2(a)(i) The kinetic energy of the car is used to do work against resistant forces such as friction on the road.

(ii) As the mass moves downwards, it loses gravitational potential energy. This loss in gravitational potential energy is converted to elastic potential energy stored in the stretched string and work done against the force exerted by the hand.

(b)(i) By conservation of energy,

Loss in KE = gain in GPE + word done against friction

$$\frac{1}{2}m(v_i^2 - v_f^2) = mgh + fd$$

$$\frac{1}{2}(55)(6.5^2 - v_f^2) = (55)(9.81)(1.3) + (2.0)(2.0)$$

$$v = 4.1 \text{ m s}^{-1}$$

(ii) If the ramp is replaced by the one shown in the figure, the distance travelled will be shorter and hence the work done against friction will be smaller. Therefore, the speed the skateboarder exits at point B will be higher.

3(a)(i) As the block oscillates in the liquid, energy is needed to overcome the